

SUPPLEMENTAL VIDEO REVIEW

Check Your Knowledge Questions and Explanations

Pharmacology of Diuretic Agents, Part 1:

- **Principles of Diuretic Action**
- **Carbonic Anhydrase Inhibitors**
- **Loop Diuretics**

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pharmacology help and discussions by appointment.**

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After completing the preparation materials, students should be able to:

1. **Sites of action:** Map each diuretic class to its site of action in the nephron and predict the downstream effects.
2. **Mechanisms:** Explain how inhibition of specific renal transporters alters sodium handling, electrochemical gradients and downstream nephron function to produce diuresis.
3. **Effects:**
 - Correlate each diuretics class with their urinary electrolyte patterns.
 - Predict acid-base disturbances based on mechanisms.
 - Explain diuretic effects on renal hemodynamics and GFR.
4. **Compensatory responses:** Relate diuretic actions to compensatory physiological responses (e.g., SNS, RAAS activation, distal nephron reabsorption) and explain their role in diuretic resistance.
5. **Therapeutic uses:** Select the appropriate diuretic based on mechanism, patient factors, and clinical scenario.
6. **Clinical considerations:**
 - Explain how the pharmacokinetic properties influence onset, efficacy, and clinical use of the diuretic agents.
 - Predict the adverse effects and contraindications based on diuretic mechanism and patient factors.
 - Predict the impact of NSAIDs on diuretic response by describing the role of the renal prostaglandins on the actions and effects of the diuretic agents.
 - Predict drug interactions based on the mechanism of action, pharmacokinetic properties, or pharmacogenetic variabilities of the individual drugs or drug classes

- The kidneys maintain plasma osmolality by filtering blood and precisely regulating water and electrolyte excretion. Coordinated tubular reabsorption and secretion of solutes and water guided by hormonal signals, including, sympathetic nervous system, RAAS, and vasopressin (ADH), fine-tune renal handling of sodium and free water to keep plasma osmolality stable.
- Many diseases distort the kidney's normal ability to balance sodium and water.
- ***The goal of diuretic therapy is to restore and maintain euvolemia by altering the kidney's regulation of sodium and water.***

To understand the actions and effects of diuretics, knowledge of the basic physiological functions of the kidney and compensatory responses to changes in extracellular volume are essential.

Preparation Materials

Required

- Dr. Goldstein's Practice Questions

Practice questions are valuable for learning because they force the brain to retrieve information, which strengthens neural pathways, interrupts the forgetting curve, and makes knowledge significantly easier to recall.

Optional materials:

Use the resources that work best for you. You don't need to study all of them.

- Lecture Videos | Dr. Goldstein's Word handout
- ScholarRx Bricks | Guided Reading Questions
- Textbook resources with links are listed on the next slide.
- Review books and ScholarRx have more practice questions.
- Write your own questions – what you would like to learn more about.

Links to resources

Scholar Rx Bricks: Renal <https://exchange.scholarrx.com/collection/renal>

Renal> Treatment of Renal Disorders > Diuretics

Diuretics Foundations and Frameworks <https://exchange.scholarrx.com/brick/diuretics-foundations-and-frameworks>

Carbonic Anhydrase Inhibitors <https://exchange.scholarrx.com/brick/carbonic-anhydrase-inhibitors>

Loop Diuretics <https://exchange.scholarrx.com/brick/loop-diuretics>

Thiazide Diuretics <https://exchange.scholarrx.com/brick/thiazide-diuretics>

Potassium-Sparing Diuretics <https://exchange.scholarrx.com/brick/potassium-sparing-diuretics>

Osmotic Diuretics <https://exchange.scholarrx.com/brick/osmotic-diuretics>

Textbooks/Exam review books you might find helpful:

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 15: Diuretic Agents

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3058§ionid=255304977>

Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 15: Diuretics and Other Drugs That Act on the Kidney

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285591587#1207298272>

LWW Health Library Premium Basic Sciences; Lippincott's Illustrated Reviews: Pharmacology 8e, 2023; Chapter 9: Diuretics

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253326102&bookid=3222>

Key Points Overview

- Diuretics classes are based on their sites of action along the nephron
- Diuretics increase Na^+ excretion and urine flow producing natriuresis and diuresis. They are used to adjust the volume or composition of body fluids in edematous states or other disorders, such as hypertension. Diuretics have effects on urine composition and renal hemodynamics. The extent of the effect is related to their sites of action.
- The three main diuretic classes in clinical use are **loop**, **thiazide (thiazide-type and thiazide-like)**, and **potassium sparing diuretics**.
- **Carbonic anhydrase inhibitors and osmotic diuretics** are clinically important but have few therapeutic uses.
- Organic acid secretory systems located in the proximal tubule secrete diuretics from the blood into the luminal side of the tubule, where most of them act except for the mineralocorticoid receptor antagonists, which diffuse through the basolateral membrane.
- Prostaglandins (PGs) contribute to the renal physiology and to the efficacy of the diuretic agents. PGE₂ and PGI₂ are the major subtypes expressed in the kidney. They cause vasodilation and blunt Na^+ reabsorption and vasopressin (ADH)-mediated water transport in collecting ducts.

Key Points Diuretic Effects

- The efficacy of a diuretic is related to several factors, including its **site** of action, **duration** of action, and **dietary salt intake**.

Diuretics can cause:

- Extracellular fluid volume contraction with compensatory responses (SNS, RAAS, ADH)
- Dehydration due to excessive water loss
- Potassium wasting or potassium sparing effect (site-dependent effects)
- Hyponatremia
- Acid-base disturbances

Stabilization with consistent diuretic therapy:

- Na^+ determines extracellular fluid volume (ECFV) Steady state: Na^+ intake = Na^+ excretion
- Initially: Natriuresis + diuresis $\rightarrow \uparrow$ urine volume and $\downarrow$ ECFV
- Compensatory mechanisms to restore Na^+ balance: $\downarrow$ ECFV $\rightarrow$ activation of homeostatic mechanisms SNS, RAAS, ADH $\rightarrow \uparrow \text{Na}^+$ reabsorption in the nonblocked upstream segments, followed by water $\rightarrow$ sustained benefit at lower baseline ECFV

Key points CAIs

- **Mechanism:** Carbonic anhydrase inhibitors (CAIs) potently inhibit both the membrane-bound and cytoplasmic forms of carbonic anhydrase and can cause nearly complete abolition of filtered bicarbonate reabsorption in the proximal tubule along with increased secretion of sodium, potassium, and water.
- **Weak efficacy:** However, sites distal to the proximal tubule collectively have a large Na^+ reabsorptive capacity. Chloride is reabsorbed in exchange for bicarbonate excretion.
- **Metabolic acidosis:** The proximal tubule is the dominant site of action of carbonic anhydrase inhibitors and the main mechanism of metabolic acidosis by CAIs. A secondary site of carbonic anhydrase inhibition in the distal nephron inhibits acid secretion and generation of new bicarbonate, which contributes to metabolic acidosis. CAIs reduce acid buffering capacity in the blood by depleting bicarbonate and, secondarily, cannot regenerate it effectively.
- **Self-limited effect:** Bicarbonate depletion reduces the amount of filtered bicarbonate, reducing drug effect. Metabolic acidosis enhances sodium reabsorption to restore sodium balance. The effects together quickly reduce CAI diuretic effect.
- **Therapeutic uses:** CAIs are used primarily in the management of open-angle glaucoma, the treatment and prophylaxis of acute mountain sickness, and metabolic alkalosis.
- Students should be aware of specific adverse effects, including potential for calcium kidney stones, and drug interactions.

Key points: Loop Diuretics

- The loop diuretics – furosemide, bumetanide, torsemide, and ethacrynic acid – are the most potent diuretics available.
- **Mechanism:** Inhibit sodium-potassium-chloride channels (NKCC2) in the thick ascending limb (TAL) of the loop of Henle and prevent sodium reabsorption, leading to natriuresis and diuresis.
- **Efficacy:** Highly efficacious because they block up to ~25% of filtered sodium reabsorption, leading to massive fluid loss.
- **Ceiling effect:** NKCC2 saturation by the loop diuretic along with sodium reabsorption in the distal nephron limits the fraction of filtered sodium blocked by loop diuretics to ~25% of filtered Na⁺.
- **Therapeutic uses:** Most commonly for patients with fluid overload.
- **Drug interactions:** High potential for drug interactions by impaired entry to nephron, electrolyte disturbances or volume depletion by other drugs, changes in sodium handling by other drugs, and additive ototoxicity.

Loop diuretics can interfere with excretion of lithium, which has a narrow therapeutic window and high potential for toxicity.

- Ethacrynic acid is a loop diuretic and the only diuretic agent that does not have a sulfonamide moiety. In the past, it may have been recommended for patients with severe allergy to sulfonamide antibiotics. Ethacrynic acid has a greater potential to cause ototoxicity than the other loop diuretics.
- Students should be aware of the numerous drug-specific adverse effects and interactions.

Principles of Diuretic Therapy

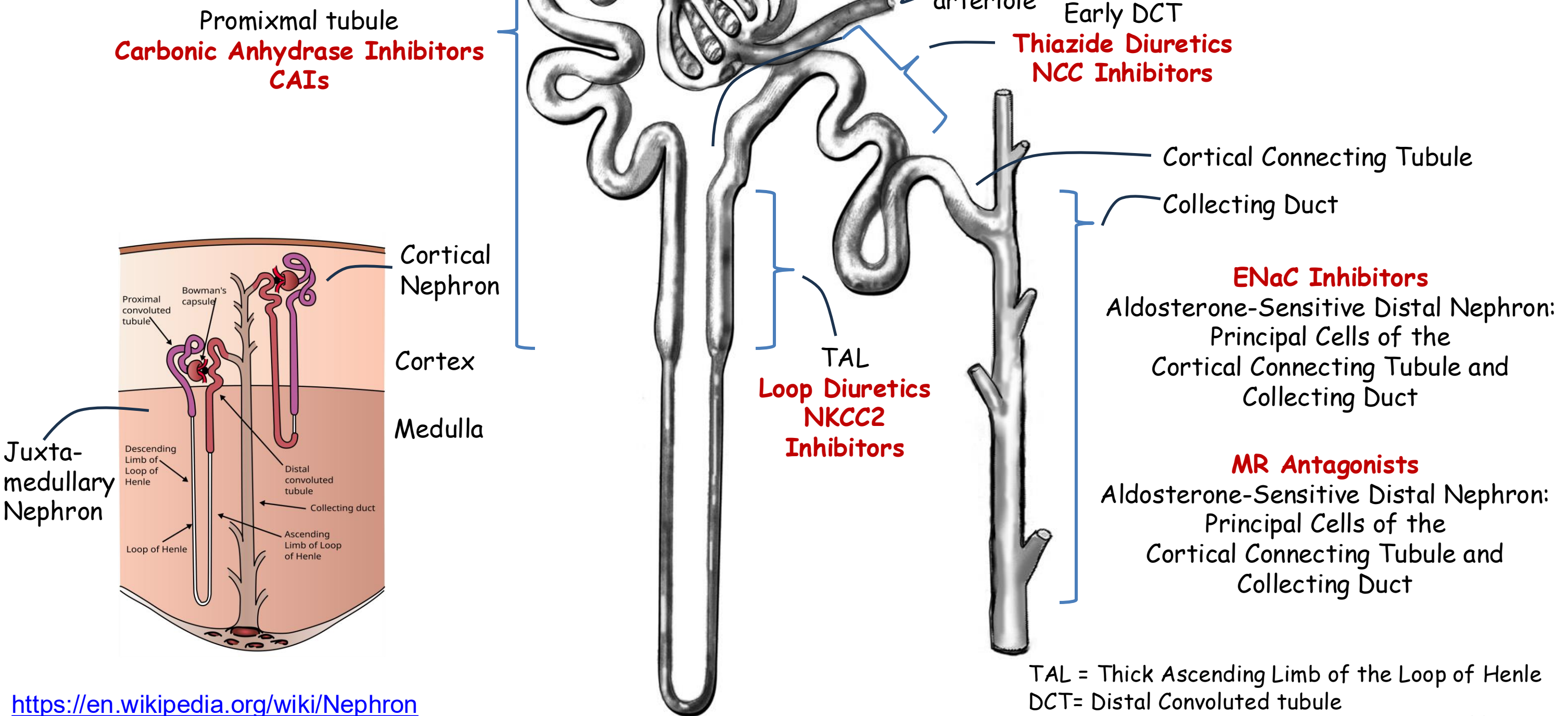
The general principles of diuretic therapy can be applied to the specific classes of diuretics and to the individual drugs in each class.

To reiterate, a solid grasp of the physiological functions of the kidney and compensatory mechanisms are **essential** for understanding diuretics pharmacology.

- Nephron structure and function
- Glomerular filtration
- Tubular reabsorption and secretion
- Electrolyte handling
- Hormonal regulation – SNS, RAAS, ADH

Please refer to physiology resources. The Notes Handout for this lecture briefly describes these functions in relation to diuretics pharmacology.

Diuretics Classes and Sites of Action



<https://en.wikipedia.org/wiki/Nephron>

Artwork by Holly Fischer

<https://www.niddk.nih.gov/news/media-library/18193>

Labels added by LG

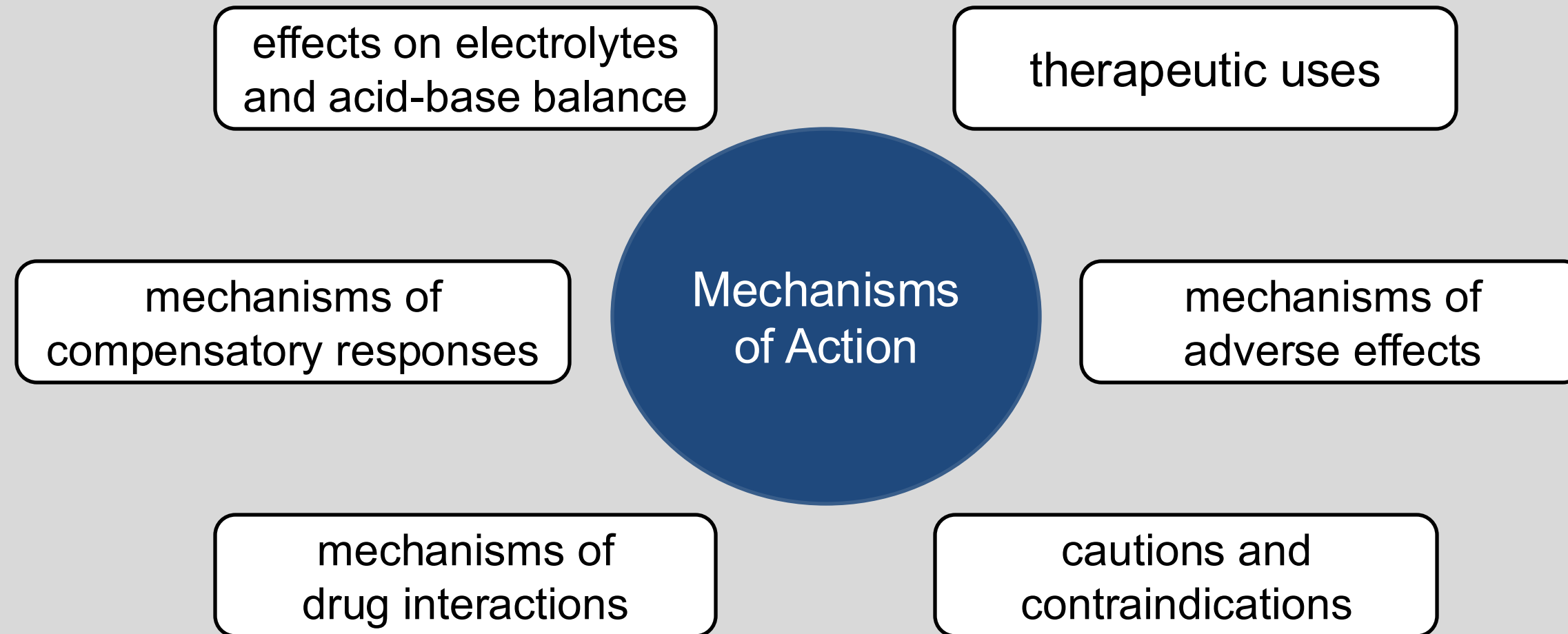
Classes of diuretics and their sites of action

Students should know the pharmacology of the starred (*) drugs by name.

Subclass	Site of Action	Drugs in the class	
Carbonic anhydrase Inhibitors (CAIs)	Proximal tubule (primary site) Collecting duct (secondary site)	*Acetazolamide, Methazolamide Dorzolamide, Brinzolamide (eye drops)	
Loop diuretics Na ⁺ /K ⁺ /2Cl ⁻ symport (NKCC2) Inhibitors	Thick ascending limb (TAL) of Henle's loop	*Furosemide *Torsemide, Bumetanide *Ethacrynic acid	
Thiazide diuretics Na ⁺ -Cl ⁻ symport (NCC) inhibitors	Distal convoluted tubule (DCT)	*Hydrochlorothiazide Chlorothiazide Methychlothiazide Bendroflumethiazide	*Chlorthalidone Indapamine Metolozone
Epithelial Na ⁺ channel (ENaC) inhibitors (K ⁺ -sparing)	Cortical collecting duct principal cells	*Amiloride Triamterene	
MR antagonists (K ⁺ -sparing)	Cortical collecting duct principal cells	*Spironolactone *Eplerenone	
Osmotic diuretics	Increase urine output by creating an osmotic gradient in the nephron	*Mannitol Urea, Glycerol, Isosorbide	

What you need to know for all classes and starred drugs

PK TAKEAWAYS that relate to patient care



- You should strive to explain why certain drugs are drugs of choice for specific disorders and others are later-line.
- You should be able to explain the potential adverse drug events that could result from drug-disease and drug-drug interactions to anticipate and mitigate potential harms and contraindications to drug administration.

Class Pharmacokinetic Properties

TAKEAWAY: Pharmacokinetics is the foundation for safe prescribing, avoiding toxicity, understanding drug interactions, and personalizing treatment for your individual patients.

	Properties	Why it matters
A	Oral, IV, IM depending on the agent and the therapeutic use Absorption differs among the agents – some have low bioavailability and others achieve excellent plasma concentrations.	High bioavailability ensures predictable delivery to the kidney and consistent diuretic effect.
D	Highly plasma protein bound, mainly to albumin → determines free (unbound) drug concentration.	Highly protein bound diuretics are not filtered but enter the nephron via secondary active transport.
	Diuretics are secreted via OAT from blood into the proximal tubule. Sites of action are on the luminal side of renal epithelial cells. <i>*Exception:</i> MR antagonists passively diffuse across basolateral membrane and bind MRs in the cytosol of the principal cells	OAT inhibitors may reduce drug concentration in nephron → may reduce diuretic efficacy.
M	Metabolism is agent specific.	Metabolism (and absorption) determines bioavailability.
E	Primarily excreted in urine as active drug.	It makes sense: The sites of action are within the nephron.
t _{1/2}	Vary by drug	Determine dosing frequency and duration of action.

Diuretics decrease ECF volume (interstitial fluid and plasma volume) and effective circulating volume, which activates physiologic compensatory responses.

1. ↑ Na⁺ excretion (natriuresis) and an accompanying anion, Cl⁻ → ↓ total-body NaCl content
2. ↑ Water output in the form of urine (diuresis) → ↓ ECF volume
3. Duration of effect and compensatory response:

Volume depletion

SNS activation + RAAS activation

rapid, supportive, limited effect *slower, dominant, sustained effect*

Aldosterone-mediated Na⁺ and water retention

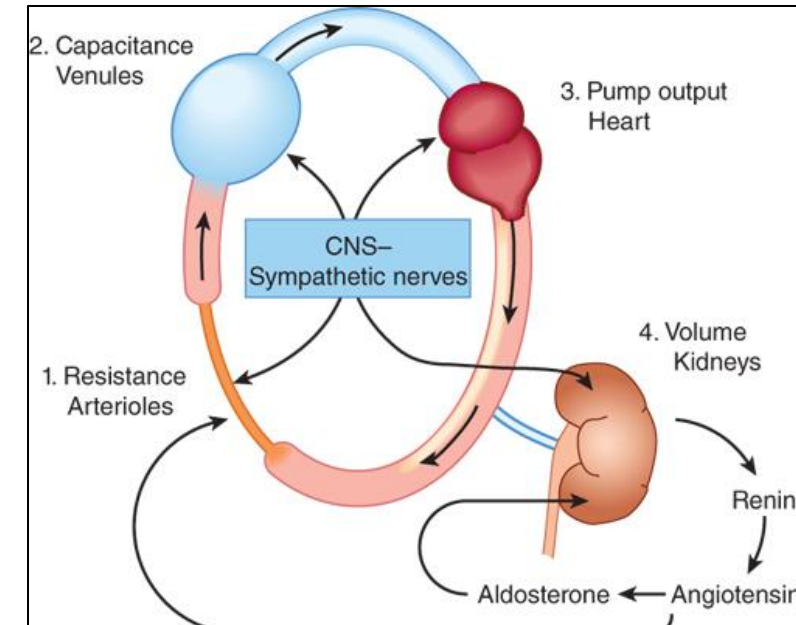
↓ ECV → baroreceptors sense volume depletion

non-osmotic ADH release

AQP2 channels are inserted in collecting duct

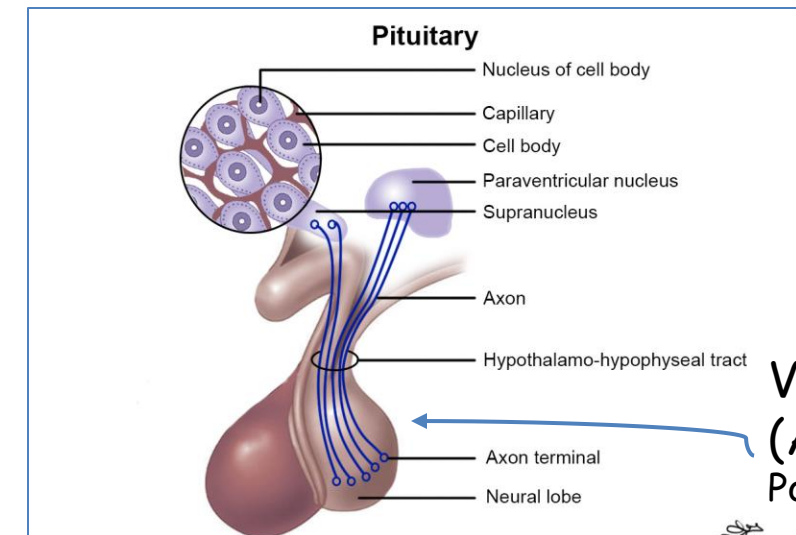
absorption of free water → ↑ effective circulating volume

Anatomic sites of blood pressure control.



Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition
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Aldosterone



**Vasopressin
(ADH)
Posterior pituitary**

<https://step1.medbullets.com/neurology/113005/posterior-pituitary>

ECV: effective circulating volume

ECF: extracellular fluid

SNS: sympathetic nervous system

RAAS: renin-angiotensin-aldosterone system

ADH: antidiuretic hormone = vasopressin



Citation: Chapter 11 Antihypertensive Agents, Todd W. Vanderah. *Katzung's Basic & Clinical Pharmacology, 16th Edition*; 2024, Figure 11-1. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281748109>
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Slide adapted by LG

Diuretics effects on electrolytes

The mechanisms are explained under the pharmacology of the specific drugs.

Diuretics modify renal handling of electrolytes:

- Hypokalemia or hyperkalemia (diuretic-specific effect)
 - K⁺ wasting effects: Diuretic-induced K⁺ (and H⁺) excretion
 - K⁺ sparing effects: Diuretic-induced K⁺ (and H⁺) reabsorption
- Hypokalemic metabolic alkalosis or hyperchloremic metabolic acidosis
- Hyponatremia
- Ca²⁺ reduced or elevated serum levels
- Hypomagnesemia
- Uric acid → hyperuricemia → ↑risk of gout
in susceptible patients

Mitigate these effects by
monitoring electrolyte
levels and
signs/symptoms

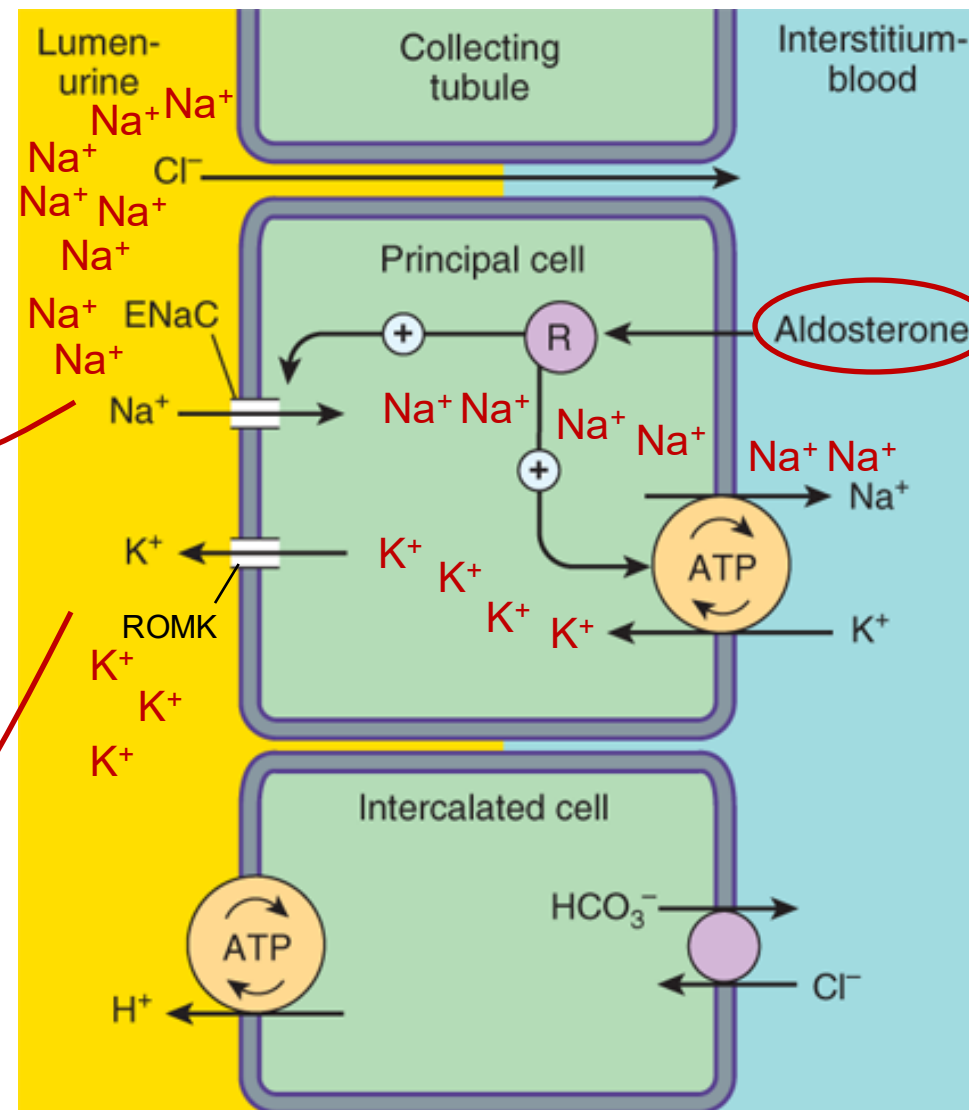
Electrolyte complications typically occur within the first 4 weeks of diuretic therapy used chronically in stable patients as the body transitions to a new steady state at a lower ECF volume.

Note: Abnormalities can occur later with dose increases, dietary Na⁺ intake, renal function decline, drug interactions.

Mechanisms of K^+ -**WASTING** in the principal cells (distal nephron)

Na^+ reabsorption is coupled with K^+ secretion.

- 1 upstream block of Na^+ transport
↓
↑ delivery of Na^+ load to principal cells in distal nephron
↓
↑ Na^+ reabsorption via ENaC (stimulated by aldosterone)
↓
↑ lumen-negative electrical gradient
↓
 K^+ moves into lumen down electrical gradient
↓
excretion in urine



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Ion transport pathways across the luminal and basolateral membranes of collecting tubule and collecting duct cells. Inward diffusion of Na^+ via the epithelial sodium channel (ENaC) leaves a lumen-negative potential, which drives reabsorption of Cl^- and efflux of K^+ . R, aldosterone receptor.

- 2 diuretic-induced volume depletion
↓
RAAS activation
↓
stimulates aldosterone-sensitive Na^+/K^+ ATPase and expression of ENaC and ROMK
↓
↑ Na^+ reabsorption
↓
↑ lumen-negative electrical gradient
↓
↑ K^+ secretion
decreases plasma K^+

Mechanisms of K⁺-**SPARING** effect in the collecting tubules (principal cells)

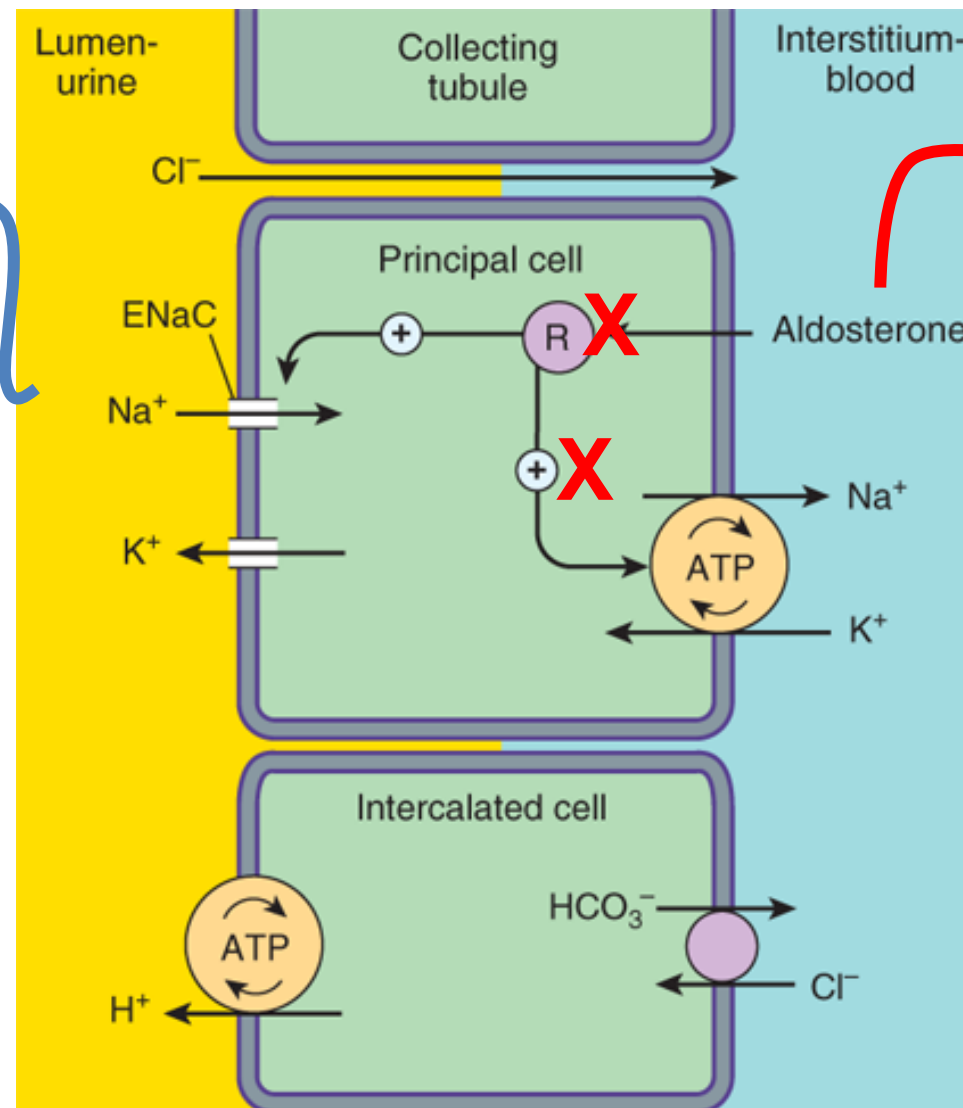
1

ENaC block increases Na⁺ in the lumen

↓
lumen negative potential diminishes

↓
K⁺ secretion stops

↓
potassium-sparing effect



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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2

MRA decreases aldosterone-mediated ENaC expression, ROMK activity, and H⁺ ATPase expression

↓ Na⁺ reabsorption → ↑ Na⁺ in the lumen and lumen negative potential diminishes

↓
K⁺ secretion is blunted

↓
potassium-sparing effect

The magnitude of effect is related to aldosterone levels.

Ion transport pathways across the luminal and basolateral membranes of collecting tubule and collecting duct cells. Inward diffusion of Na⁺ via the epithelial sodium channel (ENaC) leaves a lumen-negative potential, which drives reabsorption of Cl⁻ and efflux of K⁺. R, aldosterone receptor.

Clinical Uses of Diuretics

Edematous States

Congestive heart failure, Liver failure
Nephrotic syndrome, Acute kidney injury

Loop diuretics mobilize ECF into plasma→
kidneys→ excretion to manage volume overload

Non-Edematous States

Hypertension

Thiazides decrease PVR.

Hypercalciuria, prevent kidney stone recurrence

Thiazides decrease Ca^{2+} excretion.

Nephrogenic diabetes insipidus

Thiazides decrease urine volume.

K^{+} -sparing effect

ENaC inhibitors / MRAs

Lithium toxicity

ENaC inhibitors decrease Li^{+} accumulation in
principal cells

Hyperaldosteronism

MRAs inhibit aldosterone

Mountain sickness / Glaucoma

CAI – Acetazolamide reduces bicarbonate
availability

↑ Intracranial pressure / ↑ Intraocular pressure

Osmotic diuretic – Mannitol

Influence of Prostaglandins on Diuretic Efficacy and Inhibition by NSAIDs

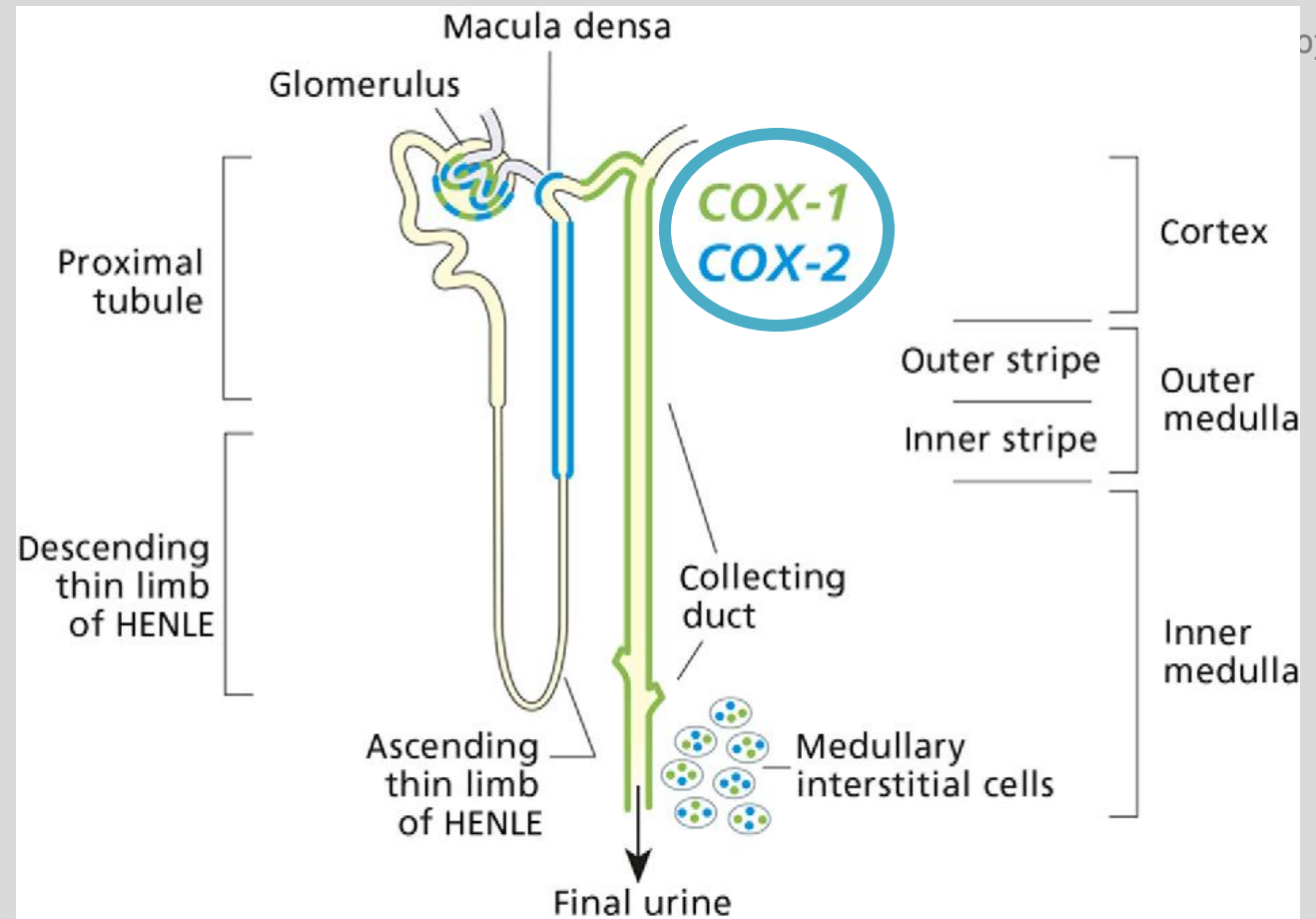
- COX-1, COX-2, and prostaglandin receptors are expressed in vascular, glomerular, and tubular compartments of the kidney.

COX-1 is constitutively expressed.

COX-2 expression is both inducible and constitutive.

- PGE2 and PGI2 are the primary bioactive prostaglandins generated in the kidney.
- PGE2 and PGI2 counteract vasoconstriction and maintain renal blood flow, glomerular filtration pressure, and GFR, support natriuresis, and promote renin release.

During states of stress, PGE2 and PGI2 are critical to preserving renal blood flow.



Nørregaard, R., Kwon, T., & Frøkiær, J. (2015). Physiology and pathophysiology of cyclooxygenase-2 and prostaglandin E2 in the kidney. *Kidney Research and Clinical Practice*, 34, 194 - 200.

NSAIDs substantially reduce the efficacy of diuretics.

- NSAIDs cause vasoconstriction of the afferent arteriole, decrease renal perfusion, reduce GFR, and cause sodium and water retention.
- NSAIDs cause edema, worsen heart failure, and worsen blood pressure in vulnerable patients.

Check your knowledge

What is the purpose of diuretic use?

What are the effects of diuretic use?

What is are compensatory effects of diuretic use?

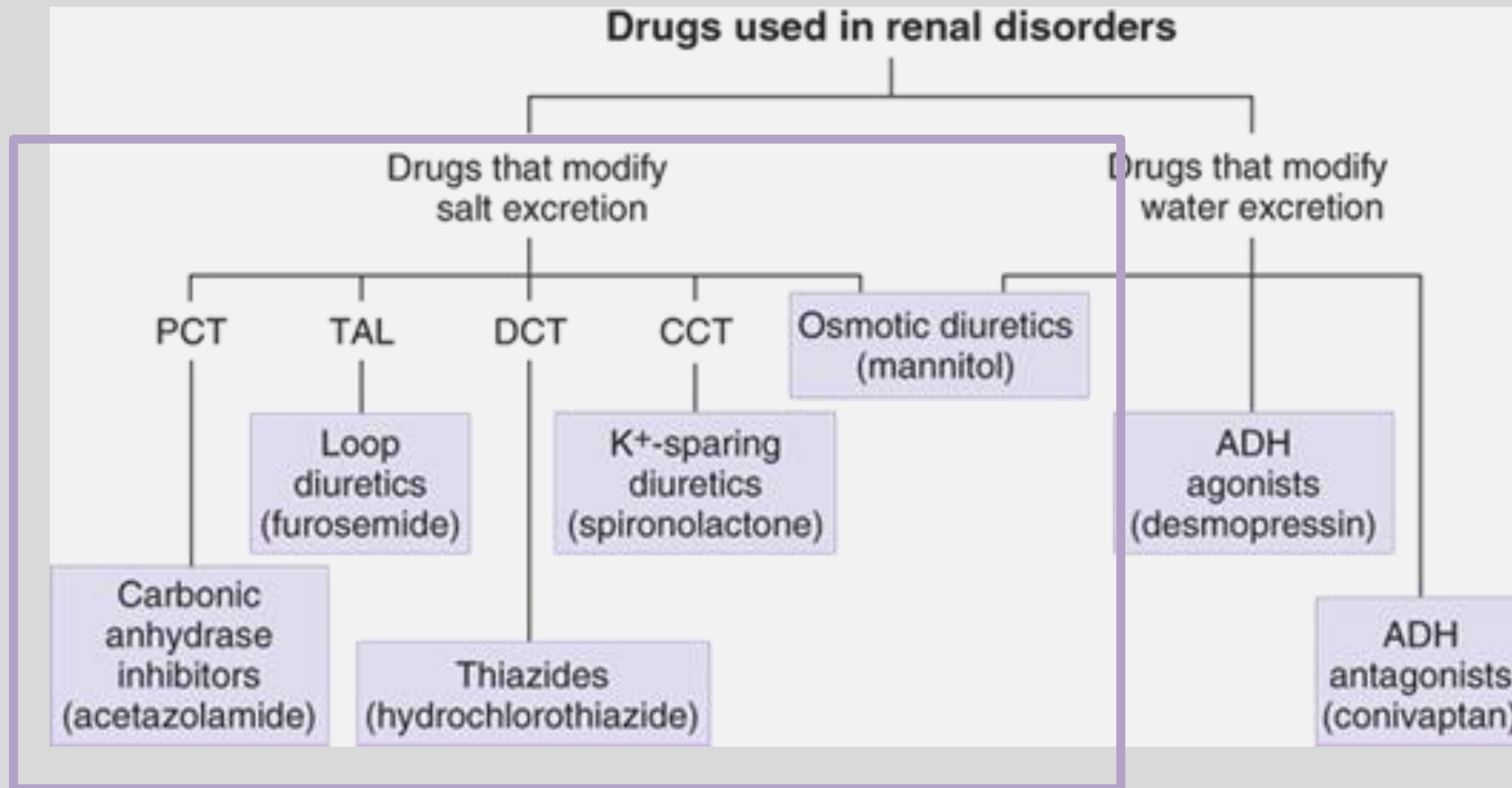
Check your knowledge

Where in the nephron do the potassium-wasting and potassium-sparing effects of diuretics occur?

What is the key determinant of potassium secretion in the distal nephron?

A 68-year-old man with chronic heart failure is treated with furosemide for worsening leg edema. After starting ibuprofen for back pain, his urine output decreases and his weight increases. **What mechanism best explains why ibuprofen reduced the effectiveness of his diuretic?**

Pharmacology of Diuretic Agents



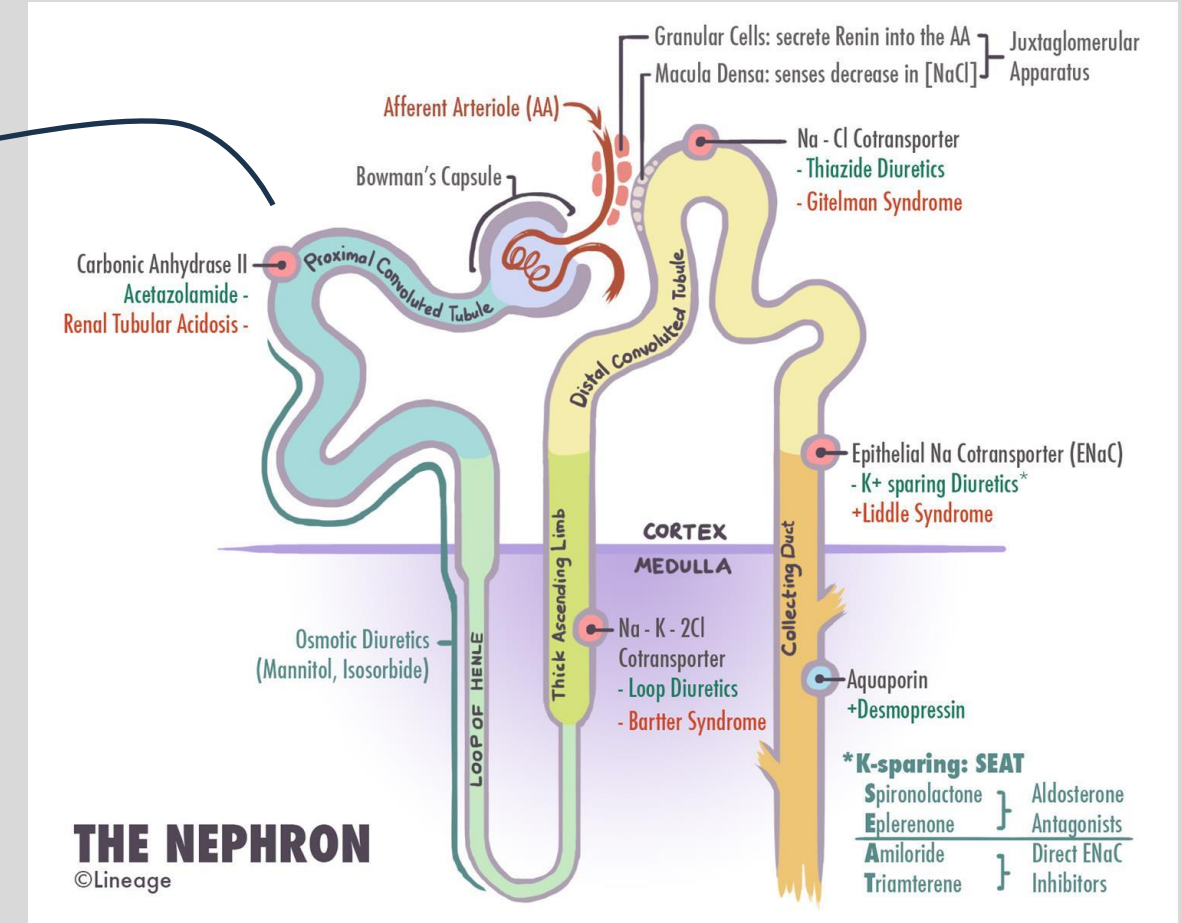
PCT: proximal convoluted tubule; TAL thick ascending loop (of Henle); DCT: distal convoluted tubule

CCT: cortical collecting tubule; ADH: antidiuretic hormone (vasopressin)

CARBONIC ANHYDRASE INHIBITORS (CAIs)

Potent, reversible inhibitors of both membrane bound and cytosolic CA in the proximal and distal tubule.

- *Acetazolamide
- Methazolamide
- Dorzolamide; Brinzolamide (eye drops)



<https://step1.medbullets.com/renal/115014/the-nephron>

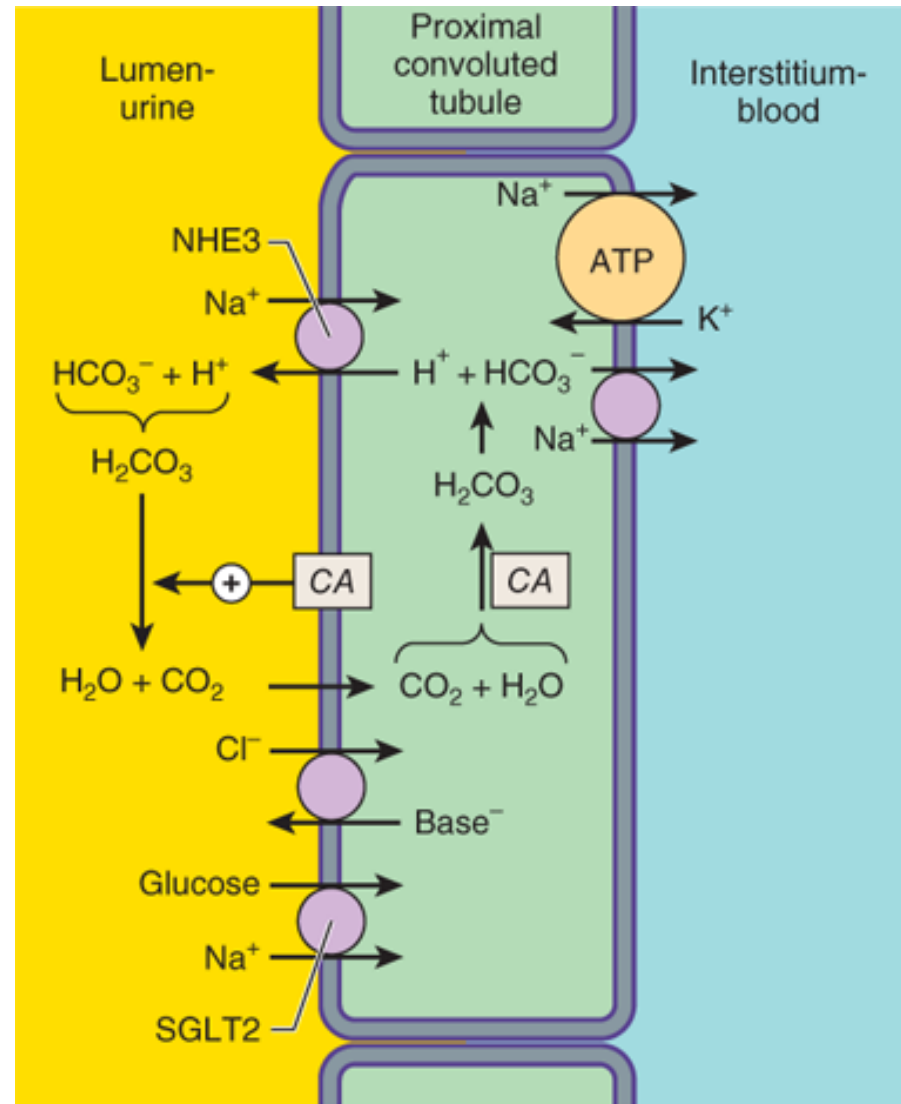
Carbonic anhydrase inhibitors block $\text{CO}_2 + \text{H}_2\text{O} \xrightleftharpoons{\text{CA}} \text{H}_2\text{CO}_3$ reactions.

CA inhibitors are potent, reversible blockers of the catalytic site of the enzyme but have **weak diuretic efficacy**.

Blocking CA in the proximal tubule blunts HCO_3^- reabsorption and impairs NHE3 activity
(NHE3: Na^+/H^+ exchanger)

↓
↓ H^+ pumped into lumen so Na^+ reabsorption decreases

↓
mild natriuresis and diuresis
with mild volume contraction



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Physiology

Na^+ and HCO_3^- are freely filtered into tubular fluid. 65-70% of filtered Na^+ and 85-90% of filtered bicarbonate is reabsorbed in the proximal tubule.

Carbonic Anhydrase Net Effect:

- H^+ in lumen combines with filtered $\text{HCO}_3^- \rightarrow \text{H}_2\text{CO}_3$.
- Lumen and intracellular CA leads to generation of intracellular H^+ .
- Basolateral Na^+ pump produces the Na^+ gradient that powers NHE3.
- H^+ generated in the cytosol is pumped into the lumen.
 - Note: NHE3 is an obligate exchanger. It requires the Na^+ gradient and intracellular H^+ to operate.

Legend: Apical membrane Na^+/H^+ exchange (via NHE3) and bicarbonate reabsorption in the proximal convoluted tubule cell. Na^+/K^+ -ATPase is present in the basolateral membrane to maintain intracellular sodium and potassium levels within the normal range. Because of rapid equilibration, concentrations of the solutes are approximately equal in the interstitial fluid and the blood. Carbonic anhydrase (CA) is found in other locations in addition to the brush border of the luminal membrane. SGLT2, $\text{Na}^+/\text{glucose}$ transporter.

CA inhibitors cause mild hyperchloremic metabolic acidosis and self-limited diuretic effect.

Metabolic acidosis:

CA inhibition leads to massive bicarbonate loss in urine.

- Continued bicarbonate loss reduces plasma bicarbonate → ↓ arterial pH

Chloride is reabsorbed to maintain electroneutrality.

hyperchloremic metabolic acidosis

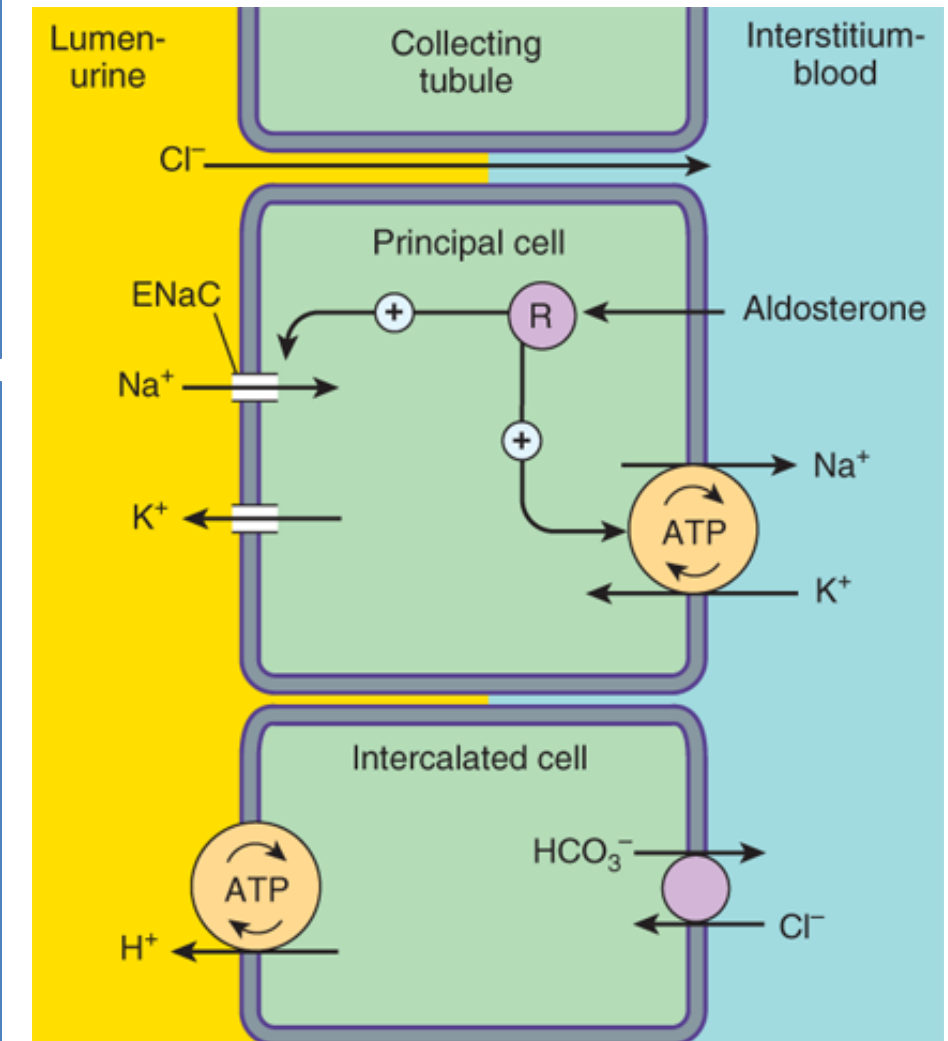
Self-limited diuretic effect (2-4 days of continuous therapy):

1. **Low filtered HCO_3^- concentration:** Large bicarbonate loss in urine causes plasma HCO_3^- levels to fall dramatically.

- Lower plasma HCO_3^- → lower filtered HCO_3^- → ↑ Na^+ reabsorption in the proximal tubule → ↓ Na^+ excretion → ↓ natriuresis and diuresis
- Cl^- is reabsorbed along with Na^+ .

2. ↓ **ECF volume + Metabolic acidosis:** RAAS / aldosterone / ADH stimulates ↑ ENaC expression + ↑ AQP2 insertion in collecting duct → ↑ Na^+ and water reabsorption

reduced diuretic efficacy



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Potassium-wasting:

↑ Na^+ reabsorption in the collecting duct → lumen becomes more negative → ↑ K^+ secretion

Slide adapted by LG

Changes in Urinary Electrolyte Patterns and Body pH in Response to CAIs				
Carbonic anhydrase inhibitors	Urinary Electrolytes:			
	NaCl	HCO3	K+	Body pH
	+	+++	+	↓
+, increase; −, decrease; 0, no change; ↓, acidosis; ↑, alkalosis Adapted from Katzung’s Basic & Clinical Pharmacology Table 15-2			potassium wasting effect	mild metabolic acidosis

RECAP: Effects of CAIs
↑ Na ⁺ excretion with water→ mild natriuresis / diuresis
↑ HCO ₃ [−] excretion → ↑urine pH / ↓ total body HCO ₃ [−] concentration → mild acidosis
↑ Cl [−] reabsorption to maintain electroneutrality → ↑ serum Cl [−] concentration → hyperchloremia
Aldosterone-mediated ↑ENaC expression and Na ⁺ reabsorption →↑ K ⁺ excretion → ↓ serum K ⁺ concentration → mild hypokalemia
CAIs reduce Ca ²⁺ tubular reabsorption / increase Ca ²⁺ excretion → may predispose patients to kidney stones. CAIs have little to no effect on Mg ²⁺ excretion.
CAIs increase solute delivery to TAL / macula densa → strong activation of TGF mechanism → afferent arteriolar vasoconstriction → ↓ GFR (sustained CAI use over time leads to TGF adaptation that blunts effect)

Clinical Applications of CA Inhibitors

- **Acute mountain sickness, prevention and treatment:**

Mild metabolic acidosis stimulates chemoreceptors and respiratory center → increases minute ventilation → increase CO₂ exhalation → ***sustained hyperventilation and improved oxygenation*** (PaO₂) → high altitude symptoms diminish

Mountain sickness results from travel to a high altitude too quickly to adjust to reduced oxygen levels. Symptoms include shortness of breath, headache, fatigue, dizziness, nausea, and insomnia. Pulmonary edema and cerebral edema can occur in severe cases.

- **Glaucoma, open-angle:**

Reducing bicarbonate production in the ciliary body reduces aqueous humor secretion, which decreases intraocular pressure for prevention of optic nerve damage.

- **Metabolic alkalosis:**

Rapid correction, especially loop diuretic-induced, which is caused by volume contraction, increased bicarbonate reabsorption, and aldosterone-stimulated H⁺ secretion.

- **Edema: *Low efficacy*** Long-term treatment is complicated by metabolic acidosis.

→ **CAIs remain highly effective in the treatment of mountain sickness** because HCO₃⁻ reabsorption is prevented and buffering capacity is depleted, and in glaucoma as bicarbonate production is inhibited.

- Only the natriuretic / diuretic effect of CAIs is self-limiting because it depends on filtered bicarbonate, which becomes depleted. (Metabolic acidosis is not self-limited.)

Adverse Effects of and Contraindications of CA Inhibitors

ADVERSE EFFECTS

- Metabolic acidosis (mild)
- Potassium wasting (mild)
- Kidney stones
 - Alkaline urine favors precipitation of calcium phosphate crystals.
- Common CNS side effects:
Paresthesia; somnolence; dizziness;
ataxia; headache; others
 - CA is widely expressed in the brain. CAls can dampen neuronal signaling.

CONTRAINDICATIONS

- Acidosis
- Severe renal disease
- Cirrhosis or other severe hepatic disease
 - CA inhibitors may contribute to hyperammonemia (NH_3) by reducing production of NH_4^+ , a main form of acid excretion.
- Hypersensitivity to the drug or any component of the formulation

Check your knowledge of carbonic anhydrase inhibitors

CAIs significantly increase the excretion of what electrolytes?

What is the effect of CAIs on the pH of blood?

How is the CAI's affect on blood pH used to therapeutic advantage?

Check your knowledge of carbonic anhydrase inhibitors

Why do CAIs cause only mild diuresis?

Why is the diuretic effect of CAIs self-limited?

Why is the action of CAIs not self-limited in the treatment of glaucoma and mountain sickness?

A 28-year-old first-time mountain climber starts acetazolamide for prevention of acute mountain sickness before ascending to high altitude. Two days later, he reports a tingling “pins and needles” sensation in his fingers and around his lips but otherwise feels well.

- 1) What is the side effect likely caused by the medication?
- 2) What is the mechanistic rationale for the medication for this purpose?

LOOP DIURETICS (NKCC2 inhibitors)

Loop diuretics block Na^+ reabsorption in the thick ascending limb (TAL) of the loop of Henle.

Loop diuretics are highly efficacious = large natriuretic effect.

- * Furosemide (sulfonamide)
- Bumetanide (sulfonamide)
- * Torsemide (sulfonamide)
- * Ethacrynic acid (phenoxyacetic acid derivative)

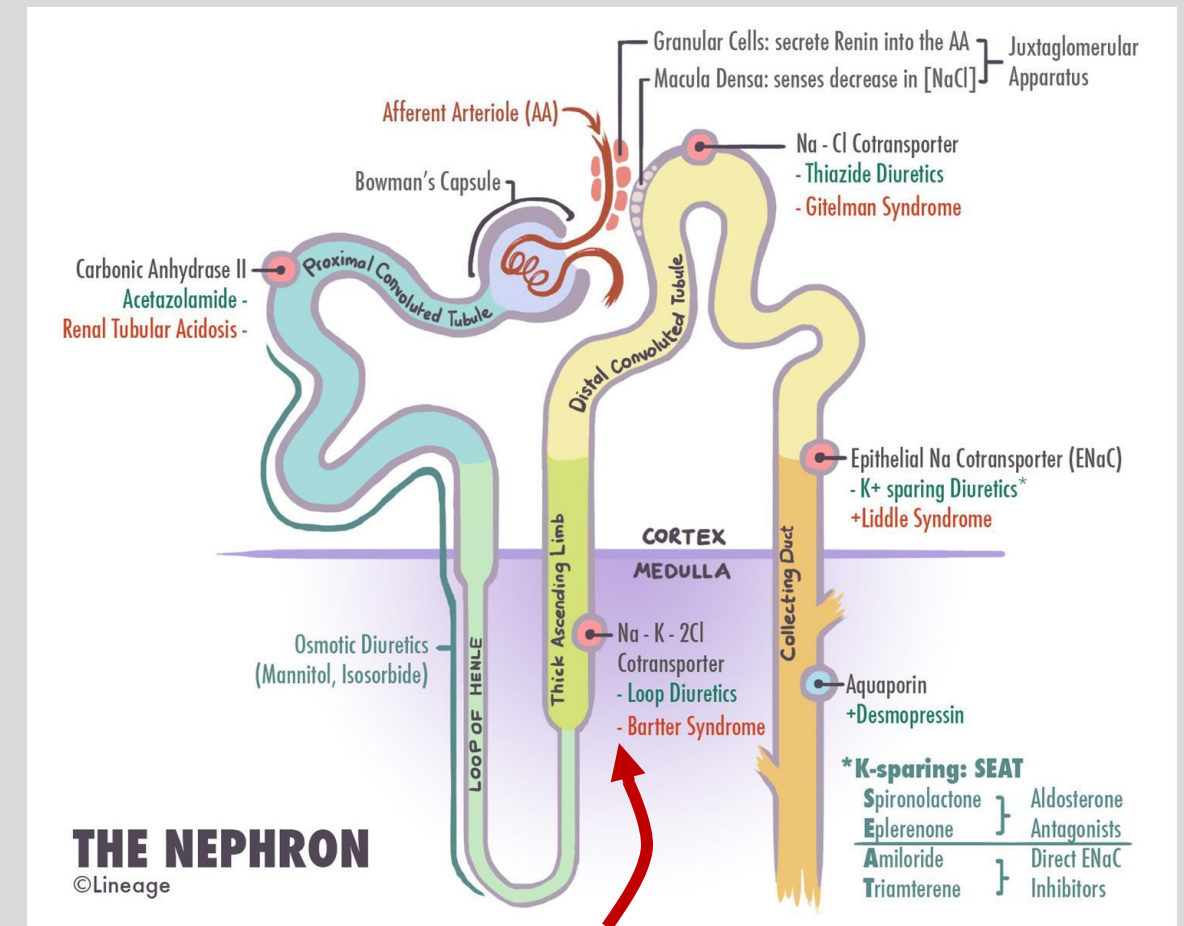
Ethacrynic acid lacks the sulfonamide moiety.

Update on immunogenic cross-reactivity potential:

The risk of cross-reactivity in persons with a history of sulfonamide allergy is clinically insignificant.

Diuretics lack the N4 aromatic amine that is the primary mechanism of the immune-mediated reaction caused by sulfonamide antibiotics.

<https://www.aaaai.org/allergist-resources/ask-the-expert/answers/2023/sulfonamide>



<https://step1.medbullets.com/renal/115014/the-nephron>

TAKEAWAYS: What to use for refractory edema and why. PK properties are typical of the class (slide 15).

Furosemide	Torsemide	Ethacrynic acid
Oral, S/L, IV infusion, subQ infusion Oral, S/L absorption ~50-60% variable absorption Why it matters: In heart failure patients, GI edema, reduced mucosal blood flow, and slowed gastric emptying can significantly reduce absorption, therefore, decrease effectiveness.	Oral only ~100% absorption and bioavailability Why it matters: In patients with refractory edema unresponsive to furosemide, it may be reasonable to switch to torsemide, which has more predictable absorption and a longer half-life than furosemide or bumetanide.	Takeaways: (boards and practice relevant) • Lacks sulfonamide moiety. It was once thought that ethacrynic acid may be preferred in patients with severe sulfonamide allergy. • Ethacrynic is NOT more effective than the other loop diuretics. • It does NOT have pharmacokinetic advantages compared to the others. • Ototoxicity risk is HIGHER, especially in combination with aminoglycosides. • Very expensive • Availability may be limited.
90-99% protein bound OAT1, OAT3 secretion Metabolized in kidney Excreted in urine t_{1/2} 1.5-2h Diuresis duration oral ~6 h, IV ~2 h	>99% protein bound OAT secretion Hepatic metabolism Excreted in urine t_{1/2} 3-5 h Diuresis duration oral ~12 h	

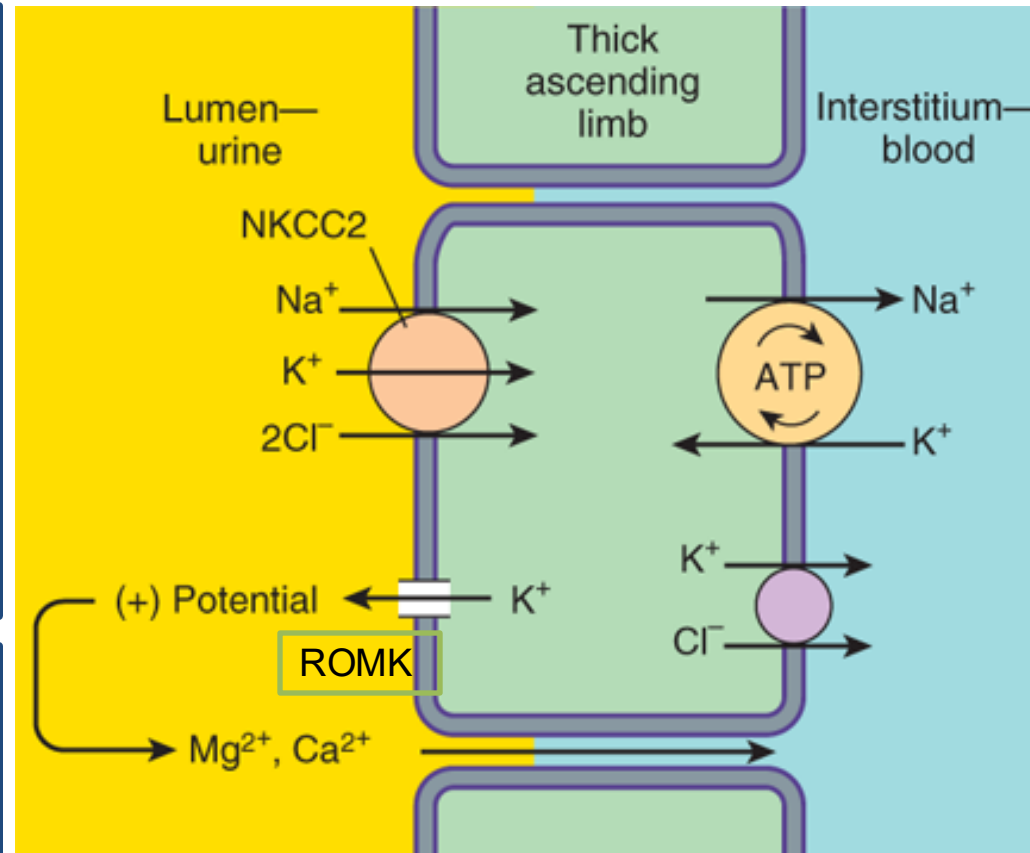
S/L: sublingual

Loop Diuretics cause profound and rapid Na^+ , Cl^- excretion and profound diuresis.

~25% of Na^+ load is normally reabsorbed in the TAL of Henle's loop.

Loop diuretics are “high efficacy” diuretics because they cause profound and rapid natriuresis and diuresis.

- Blocking NKCC2 eliminates the transepithelial potential, preventing reabsorption of Mg^{2+} and Ca^{2+} .
- Na^+ , Cl^- , K^+ , Mg^{2+} , and Ca^{2+} are excreted.



Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition
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Ion transport pathways across the luminal and basolateral membranes of the thick ascending limb cell. The lumen positive electrical potential created by K^+ back diffusion drives divalent (and monovalent) cation reabsorption via the paracellular pathway. NKCC2 is the primary transporter in the luminal membrane.

Normal function:

- Na^+ , Cl^- and K^+ enter the renal epithelial cell via the NKCC2 transporter and are reabsorbed.
- K^+ leaks back into the tubular lumen through ROMK channel, which generates a positive transepithelial potential.
- Mg^{2+} and Ca^{2+} are passively reabsorbed by paracellular transport.

ROMK:
renal outer medullary potassium channel

Physiological responses to loop diuretics (NKCC2 inhibition)

Renal Hemodynamics	Impaired Urine Concentration / Dilution	Diuretic Resistance
Acute effect: Loop diuretics stimulate intrarenal PGE₂/PGI₂ secretion → RBF increases briefly and GFR is preserved Later effect: Volume contraction → RAAS activation → mild decline in RBF and GFR	NKCC2 blockade causes: - collapse of the medullary gradient → collecting duct cannot reabsorb water effectively even with ADH present and - TAL does not dilute tubular fluid so urine diluting capacity is lost	Edema refractory to typically effective doses of loop diuretic occur due to - compensatory tubular adaptations, - renal hemodynamics, and - PK mechanisms that reduce sodium and water excretion.

Recall: PGE₂ / PGI₂ participate in local vasodilation, renin release, and salt and water transport → PGE₂ and PGI₂ **participate in the actions of loop diuretics.**

➤ ***NSAIDs reduce the efficacy of loop (and other) diuretics.***

Loop diuretics impair the kidney's ability to concentrate urine and excrete dilute urine.

Physiology:

A large amount of salt is reabsorbed via the NKCC2 in the TAL without water, which is critical for generating the hypertonic medullary interstitium and for diluting tubular fluid.

- Concentration of urine: The high osmotic gradient created in the inner medulla draws water out of the collecting ducts via AQP2 when ADH is present, which concentrates the urine.
- Dilution of urine: Reducing the solute concentration in the tubular fluid effectively dilutes the urine as it moves to the DCT. When ADH is low, dilute urine is excreted.

Pharmacology:

By blocking $\text{Na}^+\text{-Cl}^-$ reabsorption, loop diuretics impair the kidney's ability to:

- Concentrate urine during hydropenia:

Loop diuretics disrupt the countercurrent multiplier system, which diminishes or abolishes the osmotic gradient necessary for concentrating urine. Reabsorption of free water in the collecting duct via AQP2 channels is impaired, even when ADH is present and active.

- Excrete dilute urine during water diuresis with high intake of water:

Loop diuretic increase solute concentration in the tubular fluid moving into the DCT, which impairs the excretion of dilute urine below ~ 200 mOsm/kg. Kidneys lose the ability to excrete free water.

Edema refractory to typically effective doses of loop diuretic

Diuretic resistance is most prominent with loop diuretics. Only loop diuretics trigger nephron remodeling and compensatory sodium retention sufficient to produce clinically significant diuretic resistance.

Mechanisms: compensatory tubular adaptations, renal hemodynamics, and PK mechanisms

Illustrations are on next slide

Primary mechanism with chronic use:

- Distal nephron adaptation by hypertrophy of distal segments and upregulation of downstream Na^+ transport: NCC, ENaC, and activity of Na^+/K^+ ATPase → $\uparrow \text{Na}^+$ reabsorption capacity

Amplification:

- Volume depletion → Neurohumoral activation of compensatory mechanisms: RAAS activity, aldosterone secretion, and ADH actions
- Macula densa suppression → stimulates renin release

Contributing factors:

- Decreased absorption (gut edema heart failure patients) → reduced drug delivery to sites of action
- Impaired proximal tubular secretion (OATs)
- Higher sodium intake
- Drug interactions, especially NSAIDs and probenecid (OAT inhibitor)



**Diuretic
Resistance**

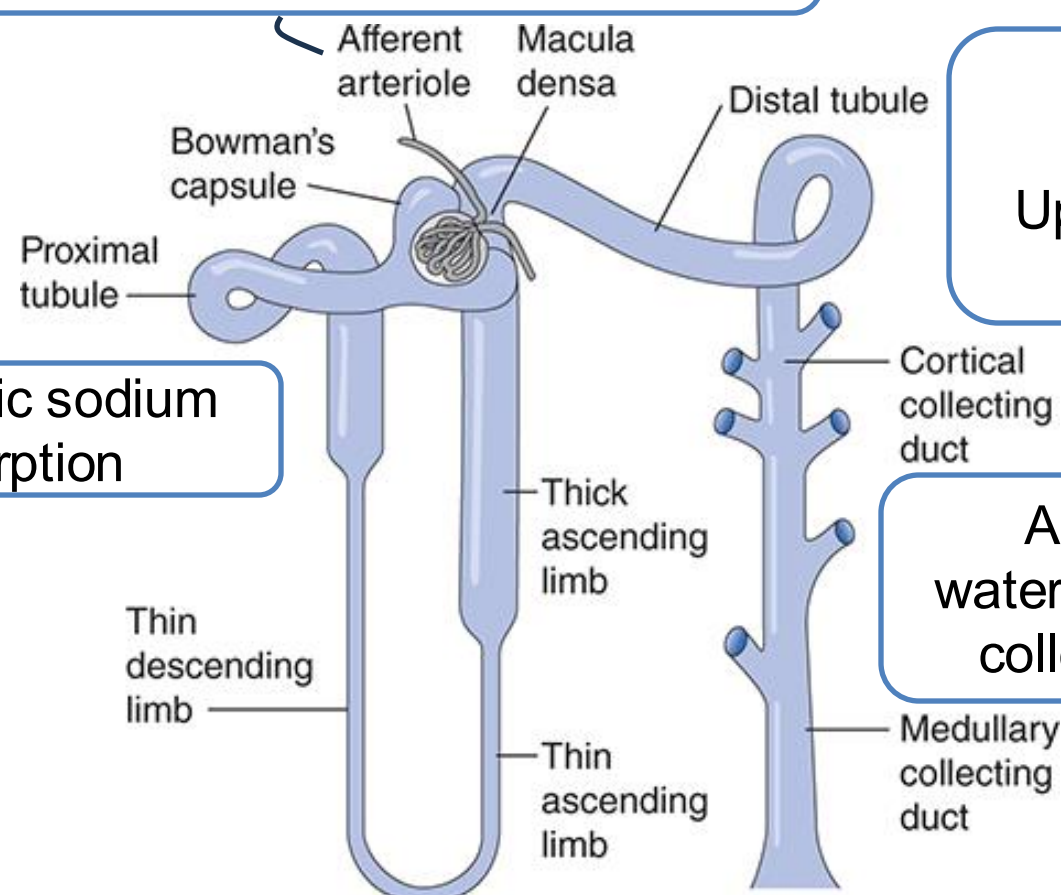
“diuretic braking
phenomenon”

Diuretic Resistance: when kidneys fail to increase sodium and fluid excretion despite maximal diuretic doses

Mechanisms: pharmacokinetic, renal hemodynamics, and compensatory tubular adaptations

PK: ↓ Absorption reduces drug delivery to nephron / short $t_{1/2}$ → limited duration

↓ renal perfusion pressure
→ ↓ drug delivery to site of action



Post-diuretic sodium reabsorption

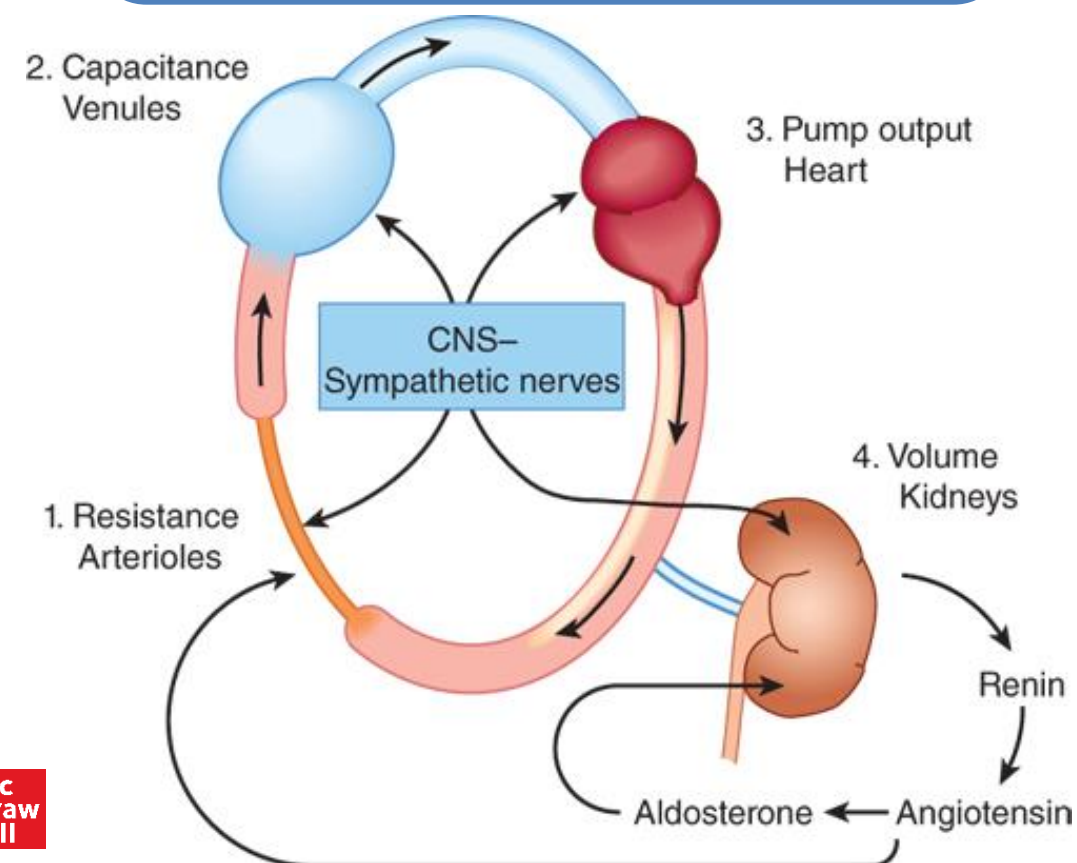
Distal nephron hypertrophy
Upregulation of Na^+ transporters

ADH-induced water reabsorption in collecting tubules



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*
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↓ ECF volume → RAAS / aldosterone
→ Na^+ /water reabsorption
and
ADH actions → water retention
and other actions



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
Copyright © McGraw-Hill Education. All rights reserved.

ECF: extracellular fluid

SNS: sympathetic nervous system

RAAS: renin-angiotensin-aldosterone system

Refractory edema: Risk factors and management tips

Refractory edema can occur in patients with:

heart failure

nephrotic syndrome

chronic kidney disease

secondary hyperaldosteronism

cirrhosis with ascites

(due to RAAS stimulation)

Apply your basic diuretics pharmacology knowledge to patient care.

Recommendations for oral management of refractory edema in patients with heart failure or substantially impaired kidney function:

Step 1: Exclude excessive salt intake and drug interactions (NSAID use; OAT inhibitor).

Step 2: Increase the dose of the loop diuretic to overcome reduced drug delivery.

Step 3: If response is inadequate, give total furosemide dose in 2-3 divided doses to reduce post-diuretic sodium retention.

Patients who have not responded to maximal oral doses of furosemide may have low GI absorption of the drug → switch to torsemide. (Torsemide has excellent absorption even in gut edema.)

Step 4: If response is inadequate, add thiazide/thiazide-like diuretic or, if hypokalemia is present, ENaC inhibitor or MR antagonist.

Changes in Urinary Electrolyte Patterns and Body pH
in Response to CA Inhibitors and Loop Diuretics

Diuretic Class	Urinary Electrolytes			
	NaCl	HCO ₃ ⁻	K ⁺	Body pH
Carbonic anhydrase inhibitors	+	+++	+	↓
Loop diuretics	++++	0	+++	↑

+, increase; −, decrease; 0, no change; ↓, acidosis; ↑, alkalosis

Adapted from Katzung’s Basic & Clinical Pharmacology Table 15-2

RECAP: Effects of Loop Diuretics

↑ Na⁺ excretion (up to 25% of filtered sodium) and water follows → profound natriuresis / diuresis

K⁺-wasting effect, ECF volume contraction and RAAS activation → hypokalemic metabolic alkalosis

Loop diuretics cause a marked increase in excretion of Ca²⁺ and Mg²⁺ due to elimination of transepithelial potential. Calcium loss is usually mild. Magnesium loss may be clinically significant.

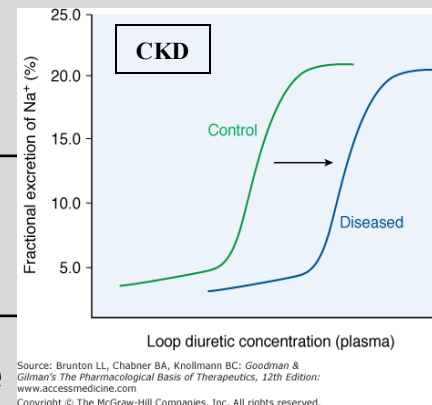
Aldosterone-mediated ↑ENaC expression and Na⁺ reabsorption → ↑ K⁺ excretion → ↓ serum K⁺ concentration → may require potassium supplementation or potassium-sparing strategy.

Loop diuretics impair the kidney’s ability to concentrate urine and excrete dilute urine.

Loop diuretics are associated with diuretic resistance.

Loop Diuretics: Clinical Applications

Clinical use	Effects and comments
Acute pulmonary edema	Natriuresis + venodilation → ↓ LV pressure and pulmonary edema
Chronic CHF	↓ Effective circulating volume minimizes venous / pulmonary congestion
Nephrotic syndrome	Relief of edema due to salt / water retention
Ascites of liver cirrhosis	Spironolactone with furosemide (oral) and salt restriction
Acute kidney injury	To relieve signs/symptoms of fluid overload Diuretic is not a replacement for dialysis.
Chronic kidney disease	Higher doses often needed (fewer nephrons, OAT competition (uremia), tubular adaptations)
Hypertension	Thiazides are a guideline-directed first-line antihypertensive Loop diuretics → risk of volume/electrolyte depletion / short duration BP lowering effect
Chemotherapy toxicity	Forced diuresis with loop diuretic and saline infusion to prevent chemo-induced nephrotoxicity <i>Potential complications of forced diuresis: rapid shifts in fluid/electrolyte balance can paradoxically lead to fluid overload, thus, pulmonary edema in addition to electrolyte disturbances, especially in patients with reduce heart, kidney, or lung function.</i>



Most are due to abnormalities of fluid and electrolyte balance.

- Hypomagnesemia (common, dose-dependent)
- Hyponatremia (milder and less frequent than with thiazides)
- Hypokalemia and metabolic alkalosis

Mechanism of hypokalemic metabolic alkalosis: Loop diuretics cause metabolic alkalosis through potassium loss and increased H⁺ secretion (via aldosterone), volume contraction, and RAAS activation that promotes proximal tubule bicarbonate reabsorption. Chloride loss with bicarbonate generation in collecting tubule maintains metabolic alkalosis.

Other toxicities:

- Ototoxicity (inhibition of NKCC1 in the inner ear disrupt endolymph K⁺ balance essential for normal hearing)
- Hyperuricemia (milder and less frequent than with thiazides)
- Hypersensitivity reactions

Pregnancy: Caution – risk of decreasing placental perfusion

Diuretics reduce maternal plasma volume and cardiac output, which can lead to decreased uteroplacental blood flow and cause electrolyte imbalances.

- Diuretics should be used cautiously if benefit to the mother outweighs risk to the fetus.
- Monitor fetal growth and maternal volume status if diuretics are used.

Drug Interactions involving Loop Diuretics

Concomitant drugs	Effect
Thiazide diuretics	Diuretic synergism can lead to profound diuresis (usually desirable)
Antihypertensive agents	Enhanced hypotensive effect (usually desirable)
ACE inhibitors / ARBs	Enhanced nephrotoxicity Volume depletion in the presence of inhibitors of angiotensin II-mediated constriction of the efferent arterioles can induce acute renal injury.
Lithium (Li^+)	Li^+ is absorbed with Na^+ mainly in the proximal tubule. Enhanced Na^+ excretion $\rightarrow$ $\uparrow$ Li^+ absorption $\rightarrow$ lithium toxicity (Loop diuretics have a smaller effect on lithium levels than thiazides.)
Digitalis glycosides	Hypokalemia $\rightarrow$ $\uparrow$ digitalis toxicity $\rightarrow$ arrhythmias
Quinidine and other QT interval prolonging drugs	Diuretic-induced hypokalemia increases the risk of torsades de pointes

Drug Interactions involving Loop Diuretics

Concomitant drugs	Effect and Mechanism
Antidiabetic agents	Impaired glucose control May involve hypokalemia wth impaired insulin release and glucose uptake → hyperglycemia
Probenecid	Competition for proximal tubule transporters → blunted diuretic response and ↑ loop diuretic's adverse effects
NSAIDs	NSAIDs ↑ Na ⁺ and water reabsorption → blunted diuretic response and can worsen metabolic alkalosis (loss of PG-mediated afferent dilation is perceived as low perfusion → RAAS activation →→→→ ↑K ⁺ , H ⁺ secretion / bicarbonate reabsorption) Prostaglandins participate in the renal actions of loop diuretics
Glucocorticoids	Increased risk of hypokalemia Glucocorticoids promote Na ⁺ retention and K ⁺ loss
Amphotericin B	Increased risk of hypokalemia Amphotericin-induced renal injury induces potassium loss
Aminoglycosides	Synergism of nephrotoxicity and ototoxicity
Cisplatin	Synergism of nephrotoxicity and ototoxicity

Check your knowledge

How does blocking the NKCC2 lead to high-efficacy diuresis?

What are the mechanisms of the loop diuretics' effect on urinary electrolytes?

Loop diuretics can cause metabolic alkalosis. What is the mechanism?

Check your knowledge

What are the contributors to edema refractory to loop diuretic therapy?

Which patients are susceptible to loop diuretic-refractory edema?

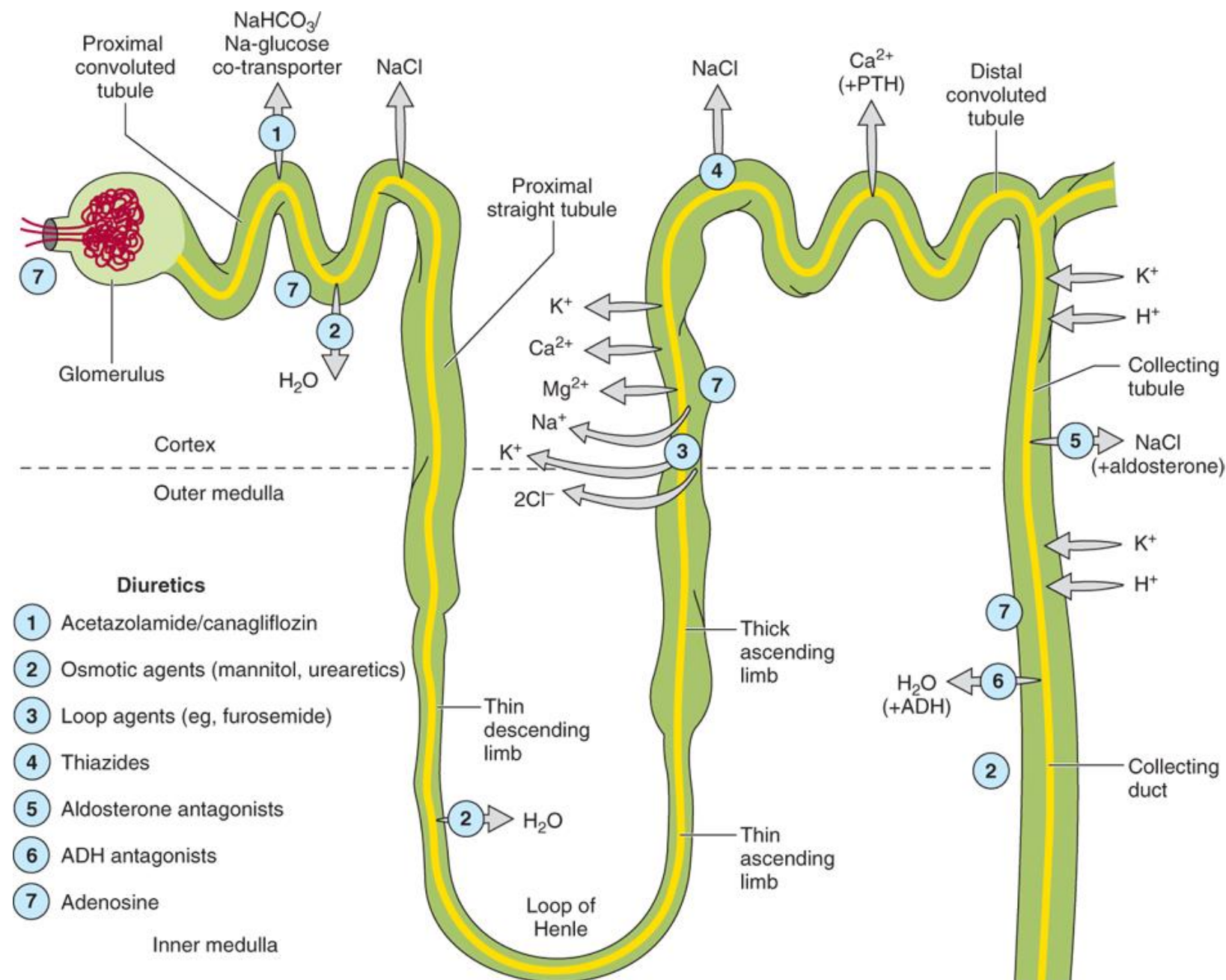
What is the mechanism by which loop diuretics impair the kidney's ability to produce dilute urine during high intake of water and concentrated urine during hydropenia?

A 75-year-old male with hypertension and heart failure is taking lisinopril (ACE inhibitor) and furosemide. After self-treating chronic knee pain with naproxen, the patient develops dizziness and a sharp rise in serum creatinine.

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

Summary

- Principles of Diuretic Action
- Carbonic Anhydrase Inhibitors
- Loop Diuretics



Tubule transport systems and sites of action of diuretics. ADH, antidiuretic hormone; PTH, parathyroid hormone.

- The goal of diuretic therapy is to restore and maintain euvolemia by altering the kidney's regulation of sodium and water. Reducing extracellular fluid volume and blood volume by stimulating natriuresis and diuresis relieves fluid overload and lowers blood pressure.
- Carbonic anhydrase inhibitors (CAIs) (eg, acetazolamide) prevent bicarbonate and sodium reabsorption in the proximal tubule. Although inhibiting the reabsorption of 65% to 70% of filtered sodium, they are mild diuretics because most of the sodium is reabsorbed in the more distal nephron segments.
- CAIs increase urinary excretion sodium, bicarbonate, potassium, and calcium. CAIs increase urine pH as bicarbonate is excreted.
- CAIs mild hyperchloremic metabolic acidosis by abolishing bicarbonate reabsorption and mild hypokalemia (potassium-wasting effect).
- CAIs are used for prophylaxis and treatment of mountain sickness by increasing ventilation, alkalosis by excretion of bicarbonate in the urine, and open-angle glaucoma by inhibiting aqueous humor secretion, which reduces intraocular pressure.

- Loop diuretics (furosemide, bumetanide, torsemide) inhibit the NKCC2 in the TAL preventing the reabsorption of ~25% of filtered sodium, which results in profound and rapid natriuresis, thus profound diuresis.
- Loop diuretics increase urinary excretion of sodium, potassium, calcium, and magnesium.
- Loop diuretics potassium-wasting effect can lead to hypokalemia.
- Loop diuretics cause metabolic alkalosis through hypokalemia, chloride depletion, and volume contraction with RAAS activation, which, together with increased H^+ loss, stimulate ammoniogenesis and impair bicarbonate excretion, which results in increased generation and retention of new bicarbonate.
- Loop diuretics are used in the treatment of edema, heart failure, and hypercalcemia.
- Loop diuretics are effective in lowering blood pressure, especially in heart failure, kidney disease, and cirrhosis. They are not first-line for management of primary hypertension because of short duration of action of adverse effects.

Answers to Review Questions

Check your knowledge of general principles

What is the purpose of diuretic use?

- To decrease extracellular fluid volume by increasing sodium and water excretion by the kidneys, with the goal of restoring and maintaining euvoemia.

What are the effects of diuretic use?

- Edema reduction: Diuretics induce natriuresis / diuresis, which promotes movement of fluid from the interstitial space into the intravascular compartment, where it can be excreted by the kidneys promotes renal excretion, thereby restoring normal effective circulating volume (euvoemia).
- Blood pressure lowering effect (thiazide diuretics): Diuretics reduce effective circulating volume by increasing renal sodium and water excretion, which lowers venous return and cardiac output initially. Over time, peripheral vascular resistance decreases → sustained reduction in blood pressure.
- Diuretics affect electrolyte balance – sodium, potassium, calcium, magnesium, bicarbonate. The

What are compensatory effects of diuretic use?

- Volume depletion activates the RAAS cascade, resulting in aldosterone-enhanced sodium and water reabsorption, counteracting the diuretic effect.
- Volume contraction → reduced plasma volume, which reduced plasma volume may stimulate secretion of vasopressin, which increases free water reabsorption in the collecting ducts.

Check your knowledge

Where in the nephron do the potassium-wasting and potassium-sparing effects of diuretics occur?

- K⁺ secretion is strongly regulated by aldosterone in principal cells in the late distal tubule and collecting duct.

What is the key determinant of potassium secretion in the distal nephron?

- Increasing Na⁺ reabsorption by ENaC creates a lumen-negative transepithelial potential that favors K⁺ secretion via ROMK.

A 68-year-old man with chronic heart failure is treated with furosemide for worsening leg edema. After starting ibuprofen for back pain, his urine output decreases and his weight increases. **What mechanism best explains why ibuprofen reduced the effectiveness of his diuretic?**

- NSAIDs block renal PGE₂ / PGI₂ synthesis → afferent vasoconstriction → ↓RBF and ↓GFR

By blocking COX-1, COX-2 and PG synthesis, NSAIDs prevent prostaglandin-mediated afferent vasodilation → reduces renal blood flow, GFR, and diuretic delivery, leading to blunted diuretic effect.

- NSAIDs inhibit OATs, reducing delivery of the drug into proximal tubular fluid.
- NSAIDs cause sodium and water retention, reducing diuretic efficacy.

Check your knowledge of carbonic anhydrase inhibitors

CAIs significantly increase the excretion of what electrolytes?

- CAIs increase urinary concentrations of **bicarbonate, sodium, potassium, and calcium**.
- Carbonic anhydrase reversibly catalyzes the forward and reverse reaction of $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3^- \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$. CA is essential for reabsorption of 80% to 90% filtered bicarbonate and generation of H^+ that drives Na^+ reabsorption.

What is the effect of CAIs on the pH of blood?

- Mild hyperchloremic metabolic acidosis
- Increased excretion of bicarbonate in the urine reduces serum bicarbonate, which is a key buffer in maintaining pH of the blood, leading to mild metabolic acidosis.
- Reabsorption of filtered chloride increases to maintain electroneutrality, which leads to a relative increase in serum chloride. Thus, CAIs cause hyperchloremic metabolic acidosis.

How is the CAI's affect on blood pH used to therapeutic advantage?

- Prevention and treatment of acute mountain sickness (acute altitude sickness).
- Mild metabolic acidosis increases stimulates ventilation, increasing CO_2 exhalation, which sustains hyperventilation and enhances oxygenation. High altitude symptoms diminish.

Check your knowledge of carbonic anhydrase inhibitors

Why do CAIs cause only mild diuresis?

- CAIs inhibit reabsorption of bicarbonate in the proximal tubule along with Na^+ .
- The nephron distal to the proximal tubule has a large Na^+ reabsorption capacity, so only a small percentage of filtered Na^+ is excreted, leading to mild natriuresis and diuresis.

Why is the diuretic effect of CAIs self-limited?

- CA inhibition leads to bicarbonate loss in urine $\rightarrow$ metabolic acidosis $\rightarrow$ over time plasma bicarbonate decreases $\rightarrow$ filtered bicarbonate load falls and, with it, Na^+ excretion is reduced.
- ECF volume contraction plus metabolic acidosis stimulates aldosterone secretion $\rightarrow$ ENaC upregulation $\rightarrow$ increased Na^+ reabsorption further limiting the natriuretic / diuretic effect.
- **Thus, diuretic efficacy is reduced.**

Why is the action of CAIs not self-limited in the treatment of glaucoma and mountain sickness?

- Acute mountain sickness treatment is due to induction of metabolic acidosis.

Metabolic acidosis is not a self-limited effect because HCO_3^- reabsorption continues to be prevented and buffering capacity of the blood is depleted.

- Glaucoma treatment goal is reduction of intraocular pressure by inhibiting bicarbonate formation, which reduces aqueous humor production.

A 28-year-old first-time mountain climber starts acetazolamide for prevention of acute mountain sickness before ascending to high altitude. Two days later, he reports a tingling “pins and needles” sensation in his fingers and around his lips but otherwise feels well.

- 1) What is the side effect likely caused by the medication?
- 2) What is the mechanistic rationale for the medication for this purpose?
 - 1) paresthesia
 - 2) Mechanism:

With rapid ascent to high elevation (~8200 ft) the body does not have time to adjust to the lower oxygen pressure, which can cause headache, nausea/vomiting, fatigue, dizziness, sleep disturbance.

The body responds to hypoxia sensed by peripheral chemoreceptors, resulting in hyperventilation. CO₂ exhalation leads to hypocapnia, which leads to respiratory alkalosis, which blunts peripheral chemoreceptors and inhibits the respiratory center, limiting further hyperventilation.

- Inhibiting carbonic anhydrase causes mild metabolic acidosis, which stimulates peripheral chemoreceptors and respiratory center → increases minute ventilation → ***sustained hyperventilation and improved oxygenation*** (PaO₂)

High altitude symptoms diminish.

Check your knowledge of loop diuretics

How does blocking the NKCC2 lead to high-efficacy diuresis?

- ~25% of Na^+ reabsorption occurs in the TAL via the NKCC2. Blocking this transporter results in profound and rapid excretion the sodium load, which results in profound diuresis.

What are the mechanisms of the loop diuretics' effect on urinary electrolytes?

- Inhibition of NKCC2 by loop diuretics prevents reabsorption of sodium, chloride and potassium and eliminates the positive lumen potential preventing paracellular transport of Ca^{2+} and Mg^{2+} . Therefore, sodium, chloride, potassium, calcium and magnesium are excreted in urine.

Loop diuretics can cause metabolic alkalosis. What is the mechanism?

- Loop diuretics cause metabolic alkalosis through potassium loss and increased H^+ secretion (regulated by aldosterone), volume contraction, and RAAS activation that promotes proximal tubule bicarbonate reabsorption. Hypochloremia impairs bicarbonate excretion and increased H^+ secretion generates and retains bicarbonate, maintaining loop diuretic-induced alkalosis.

Check more of your knowledge about loop diuretics

What are the contributors to edema refractory to loop diuretic therapy?

- Compensatory sodium reabsorption occurs after the drug wears off.
- Distal nephron hypertrophy and upregulation of sodium transporters in distal tubule enhances Na^+ reabsorption.
- Volume contraction can lead to reduced renal perfusion, which decreases drug concentration at the site of action.
- Persistent activation of SNS and RAAS leads to aldosterone-mediated $\uparrow \text{Na}^+$ and water reabsorption.
- Decreased blood volume and increased plasma osmolality stimulates vasopressin-induced water reabsorption in collecting tubules, esp. in patients with heart failure, cirrhosis, or nephrotic syndrome.

Which patients are susceptible to loop diuretic-refractory edema?

- Patients with heart failure, chronic kidney disease, liver cirrhosis with ascites, nephrotic syndrome, secondary hyperaldosteronism.

What is the mechanism by which loop diuretics impair the kidney's ability to produce dilute urine during high intake of water and concentrated urine during hydropenia?

- Normally, TAL absorbs solutes without water (TAL is impermeable to water), which dilutes the tubular fluid.
- Blocking NKCC2 disables the diluting segment so it cannot maximally dilute urine even when water intake is high.
- Loop diuretics abolish the medullary osmotic gradient so water reabsorption is limited. Urine cannot be concentrated when water intake is low (hydropenia) even when vasopressin is present.

A 75-year-old male with hypertension and heart failure is taking lisinopril (ACE inhibitor) and furosemide. After self-treating chronic knee pain with naproxen, the patient develops dizziness and a sharp rise in serum creatinine.

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

Triple hit:

Blocking prostaglandin biosynthesis + blocking ANG II effects + ↓ effective circulating volume leads to collapse of glomerular filtration pressure and GFR → risk of acute kidney injury

1. NSAIDs block prostaglandin-mediated afferent dilation, leading to afferent constriction, decreased glomerular filtration pressure and GFR → risk of acute kidney injury.
2. ACE inhibitors/ARBs block angiotensin II-mediated efferent constriction, leading to efferent dilation, decreased glomerular filtration pressure and GFR → risk of acute kidney injury.
3. Diuretics cause ECV depletion, leading to reduced effective circulating volume, leading to decreased renal blood flow, decreased glomerular filtration pressure and GFR → risk of acute kidney injury.

ECV depletion activates RAAS but RAS inhibitors (ACE inhibitors/ARBs) block ANG II effects.



Diuretic therapy can dramatically improve quality of life.

Resources

- Access Medicine Goodman & Gilman The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 29: Regulation of Renal Function and Vascular Volume
- Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 15: Diuretic Agents
- Clinical Key Boron & Boulpaep, Medical Physiology, 3e, 2017; Chapter 39: Transport of Acids and Bases
- UpToDate Lexidrug monographs

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4RR>